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BME300: Critical Analysis Report

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Genetically Engineering Probiotic Neoantigen Vectors for Tumor-Specific Immunotherapy: Critical Analysis Report

Background

Genetic engineering techniques in cancer therapy have advanced significantly over the past few decades, from cytotoxic chemotherapeutics to more personalized immunotherapies that selectively target tumours. An example of a new technology is the use of tumour neoantigens, which are peptide fragments derived from tumour-specific somatic mutations. These neoantigens are expressed exclusively by the tumour, making them ideal targets in precise targeting immunotherapy. This benefit reduces the risk of autoimmunity, and by contributing to this platform to target cancer, synthetically engineered probiotic bacteria can act as a delivery system for tumour-specific neoantigens.

Redenti et al. 's genetic engineering techniques and experimental design involved the development of probiotic neoantigen-delivery vectors for precision immunotherapy [1]. They engineered the Escherichia coli strain Nissle 1917 (EcN), a probiotic strain with an established ability to colonise tumours and act as an immunostimulatory [1,2]. This natural detection and accumulation of bacteria is a well-documented feature of bacterial cancer therapies, due to the pathogen-associated molecular patterns (PAMPs) and other immunostimulatory signals [3]. Therefore, programming EcN allows them to continuously produce arrays of tumour-specific neoantigens that will prime antigen-presenting cells (APCs) and to deliver a platform to combat the abnormal vasculature and immunosuppressive microenvironment of tumours.

Through the use of EcN, ecoli mechanisms that we already know of can be leveraged to expand our use of genetic engineering in microbiome environments. Traditional neoantigen vaccination strategies, such as synthetic long peptide (SLP) vaccines or mRNA-based vaccines, have the limitation of producing complete tumour regression in patients. These technologies are dependent on the quality of administration and the mechanisms of antigen exposure, lacking the flexibility to adapt to different tumour microenvironments (TME). Whereas bacterial vectors are able to colonise the TME directly and stimulate the immune system due to their foreign properties [3]. Biologically, it is known that bacteria selectively accumulate in tumors after injection. This allows them to intervene intravenously with tumor development and ensure that other cells are not targeted in the body. Tumour neoantigens are present only in specific areas, minimising risk for autoimmunity in cancer treatment [4]. Their work would be more efficient than traditional neoantigen vaccines due to their long-lasting effectiveness. Redenti et al. build on these concepts by leveraging the intrinsic immunogenicity to instruct APCs in the precise locations and overcome this limitation of conventional neoantigen vaccines and microbial immunotherapy platforms.

The concept of neo-epitopes, short MHC-binding peptides derived from neoantigens that display on APCs to be recognized by T-cell receptors that trigger a tumour-specific immune response, is important to the biological mechanisms that drive their research project. MHC I epitopes, activating cytotoxic CD8+ T cells, and MHC II epitopes, stimulating helper CD4+ T-cells, are incorporated into the development of verifying the neoantigen platform they develop works. The engineered bacteria are able to continuously produce the antigens that are important for continued reporting, activation, and cytosolic antigen delivery and T-cell priming by MHC presentation to activate CD8+ cytotoxic T cells and CD4+ helper T cells [5,6]. They also involve the concept of agretopicity to confirm the selection of neoantigens likely to break tolerance, create a robust immune response while not causing immunological complications to nontarget areas.

Methodologies and Approach Taken

The engineering methodologies started with their prototype synthetic neoantigen construct, assembled by concatenating long peptides linking CD4+ and CD8+ T-cell epitopes derived from the CT26 mutanome colorectal carcinoma neoantigens [7]. The peptides were predicted computationally by sequencing CT26 tumours using established bioinformatic pipelines to identify the highly expressed tumour specific mutations and ranking each of their predicted MHC binding affinities. This would allow for more efficient stimulation of cellular immunity by allowing endogenous processing and presentation [8]. However, initial expression of this prototype in wild-type EcN was low across multiple promoters.

They followed by incorporating 5-mer glycine-serine linkers between neoantigen long peptides to enhance immune targeting of arrays. Flexible glycine-serine linkers prevent steric hinderance between adjacent peptide sequences and improve folding, but too long or short variants did not improve structural benefits. Additional notable genetic engineering techniques they used to make their mutant recombinant EcN were removing cryptic plasmids in order to prevent supression in the copy number of transformed recombinant plasmids. Through removal of Lon and OmpT intracellular proteases, they were able to increase production of synthetic neoantigen constructions 80-folds higher. Some consequences of removal of these proteases were greater susceptibility to phagocytosis by bone marrow-derived macrophages (BMDMs), increasing uptake by APCs and greater sensitivity to clearance in human blood, reducing the risk of systemic bacteria [9]. The final editing they did was incorporating listeriolysin O (LLO), a pH sensitive pore-forming toxin that enables the bacteria engulfed by the phagocytes to escape from the phagolysosome into the cytosl. By expressing LLO constitutively, there is increased cytosolic delivery of the neoantigens of interest, which promote the Major Histocompatibility Complex (MHC) class I antigen presentation pathway and activate cytotoxic CD8+ T-cells [10]. In addition, they removed proteases dION to prevent autoimmunity or degradation of the neoantigens that will be expressed in their engineered EcN.

Key Data and Supportive Control Data

Their in vitro experiments systematically validate each engineering modification done before in vivo testing with mouse models. For instance, they observed low baseline expression from the prototype construct, but with GS linkers to improve expression, cryptic plasmid removal, protease deletions, and LLO to enable cytosolic antigen delivery using immunoblot, ELSIA, and assay data, they assert significant improvements to validate their synthetic EcN construct.

In their CT26 colorectal model, they intratumourally injected the EcNcΔlon/ΔompT/LLO n^{Ag19} EcN to achieve complete response of 3 out of 7 tumours. Whereas wild-type EcN provided no significant therapeutic benefit. This demonstrates that the genetic engineering modifications made are responsible for the therapeutic effect. The intravenous administration in the CT26 model conveys significant tumour growth suppression and extended survival, with detection of engineered microbes at high density and selectivity at the targeted tumour location. Furthermore, their treatment produced 100% survival within 45 days post-engrafment, while there were zero survivors in the PBS and no-neonatigen control groups.

In the B16F10 melanoma model a more aggressive cancer variant and immunosupressive than the CT26 that would require a lot of flexibility and tumor-specific targeting to control, they developed a nAg42 type recombinant that demonstrated 72% survival at 50 days post-engrafment with zero survivors in the control groups PBS, no-neoantigen, by day 24 to 30. This supports their platform that EcNcΔlon/ΔompT/LLO n^{Ag42} showed significantly more cDC1s, cDC2s, conventional CD4+ and CD8+ T-cells, natural killer (NK) cells, and inflammatory monocytes, along with increased expression of Granzyme-B and IFNγ. Additionally, there were reductions in regulatory T-cells, MDSCs, MHC-IIlo macrophages, and TIM-1+ regulatory B-cells. Depletion experiments confirm that both CD4+ and CD8+ T-cells are required for efficacy, consistent with the MHC class I and II targeting design of the neoantigen constructs.

Their methods of immunofluorescence microscopy confirms that LLO expression enabled recombinant antigens to escape from phagolysosomes and into the cytosol of BMDMs. This followed with enhanced IL-1270 secretion by recruited dendritic cells and promotes TH-1 type immunity [11]. The outcome was expected as LLO is known to act as a recognition reception agonist and immune activator.

The immunological memory strengthens the platform's immune system. The treatment with EcNcΔlon/ΔompT/LLO n^{Ag19} shows significantly reduced tumour growth, and with ex-vivo restimulation with the bacterially encoded neoantigens, there is confirmed the presence of neoantigen specific IFNγ secreting CD4+ and CD8+ TILs in treated mice only. Purified TILs from nAg19-treated mice also specifically killed CT26 but not 4T1 tumour cells, showing the tumour-specific cytotoxicity of their plaform.

Summary of Discussion

The authous' concluded that their engineered probiotic bacterial vector achieves efficient cytosolic delivery of tumour neoantigens, and it is well supported by the given data. Immunofluoresecence microscopy confirmed LLO-mediated phagolysosomal escape, and the presence of His-tagged neoantigen constructs in both tumour and tumour-draining lymph node lysates after intravenous administration confirmed that bacteria encoded neoantigens produced the relevant genes of interest in vivo. The 80-fold increase in neoantigen production relative to wild-type EcN by removal of cryptical plasmids and protease deletion addresed the fundamental limitations of bacterial vaccine platforms that have been constrained by low antigen yields. The constittuive promoter instead of an inducible system simplifies the clinical manufacturing and administration of their product. However, this also indicates that antigen production cannot be modulated after administration which may be a potential limitation.

IFNγ and IL-2 ELISA data, CSFE proliferation assays, flow cytometric detection of IFNγ+ TILs after neoantigen peptide restimulation, Granzyme-B expression analysis, and tumour cell specific cytotoxicity assays collectively demonstrate that their platform acts a robust and antigen-specific cellular immune response. Additionally, their biodistribution data demonstrate that the EcNcΔlon/ΔompT/LLO+ selectively colonises tumours at high density while clearing from other organs across both subcutaneous and metastatic models in different genetic backgrounds. This selectivity is associated with the established trait of preferential bacterial accumulation in tumours driven by abnormal vasculature, hypoxia, and an immunosuprresive microenvironment that impaoirs bacterial clearance. The authors note that there is no difference in tumour colonisation efficiency observed between their modified EcNcΔlon/ΔompT strain and wild type EcN.

The study also emphasizes the potential adaptation to different tumour models by redesigning neoantigen constructions for B16F10 and CT26. The adaptability underscores how this platform has the potential in personalization by using similar algorithmic pipelines for the neoantigens [12]. This proof of concept supports the papers claim that it is modular and tumour agnostic, showing clinical benefits of personalized neoantigen vaccienes dependent on the cost of the bioinformatics and manufacturing pipeline.

Analysis of the Conclusion

The authors acknowledge that a critical challenge is the absence of a standarized algorithm to predict mutations in immunogenic epitopes. Although there is a multi-criteria selection strategy that incorporates MHC binding affinity prediction, agretopicity assessment, variant allele fractions, and gene expression levels, their is the potential of false positives. Low-affinity neoantigens can be immunogenic, but this risks diluting the antigen pool with non-immunogenic peptides that could compete for APC processing and MHC presentation that do not produce as effective T-cell response. Additionally, there is no specific of timing, magnitude, and duration of the antigen production with administration. In clinical settings, there may be different tumour microenvironments and immune baseline, where there is insufficient antigen that primes T-cell response or excessive antigen, causing T-cell exhaustion.

Large-scale manufacturing of personalized bacterial strains has many limitations as each patient’s tumour requires de novo exome transcriptome sequencing, bioinofrmatic, neoantigen prediction, synthesis and cloning of personalised neoantigen constructs, and quality control of the resulting therapeutic strains. This must accommodate to the timeline of disease progression, which could be difficult with cancers such as B16F10 melanoma that progress rapidly. The immunogenicity of predicted human neoantigens is also more difficult to validate prospectively compared to murine models, especially with absence of well-established in vitro primitive systems [13,14]. There is still the gap of proof of concepts to human clinical translation.

In future applications, there can be combination with other immunotherapy modalities such as CAR-T cell therapy. Also there can be the risk of horizontal gene transfer, where neoantigen-encoding plasmids or resistance markers are transferred to regular bacteria, requiring more assessment and advanced safety features. The ability of EcNcΔlon/ΔompT/LLO+ nAg19 to prevent tumour growth with additional exposure suggests that there is durable immunological memory. This could translate into long-term protection against tumour recurrence following therapy. Future studies should assess this memory response over extensive periods and evaluate different strategies in dosages.

Redenti et al.'s paper represents a significant contribution to the immunotherapy field, demonstrating that the engineered probiotic bacterium can serve as a effective, safe, and adaptable delivery mechanism to personalize tumour neoantigen arrays. There was enhanced neoantigen production, improved immunological properties, and safety in their models with cryptic plasmid removal, Lon and OmpT protease deletion, 5-mer GS linkers, and LLO expression. However, some challenges remain before clinical evaluations such as neoantigen prediction algorithms requiruing further validation and standardization. Manufacturing and scalability of personalized bacteria therapeutics are difficult with more complex human tumours. If these challenges can be addressed, the probiotic neoantigen delivery platform Redenti et al. developed, can transform the approach to precision cancer immunotherapy.

References

1. Redenti, A., Im, J., Redenti, B., Li, F., Rouanne, M., Sheng, Z., Sun, W., Gurbatri, C. R., Huang, S., Komaranchath, M., Jang, Y., Hahn, J., Ballister, E. R., Vincent, R. L., Vardoshivilli, A., Danino, T., & Arpaia, N. (2024). Probiotic neoantigen delivery vectors for precision cancer immunotherapy. Nature, 635(8038), 453–461. https://doi.org/10.1038/s41586-024-08033-4
2. Canale, F. P., Basso, C., Antonini, G., Perotti, M., Li, N., Sokolovska, A., Neumann, J., James, M. J., Geiger, S., Jin, W., Theurillat, J. P., West, K. A., Leventhal, D. S., Lora, J. M., Sallusto, F., & Geiger, R. (2021). Metabolic modulation of tumours with engineered bacteria for immunotherapy. Nature, 598(7882), 662–666. https://doi.org/10.1038/s41586-021-04003-2

3. Iwasaki, A., & Medzhitov, R. (2015). Control of adaptive immunity by the innate immune system. Nature immunology, 16(4), 343–353. https://doi.org/10.1038/ni.3123
4. Schumacher, T. N., & Schreiber, R. D. (2015). Neoantigens in cancer immunotherapy. Science (New York, N.Y.), 348(6230), 69–74. https://doi.org/10.1126/science.aaa4971
5. Dingjiacheng Jia, Qiwen Wang, Yadong Qi, Yao Jiang, Jiamin He, Yifeng Lin, Yong Sun, Jilei Xu, Wenwen Chen, Lina Fan, Ruochen Yan, Wang Zhang, Guohong Ren, Chaochao Xu, Qiwei Ge, Lan Wang, Wei Liu, Fei Xu, Pin Wu, Yuhao Wang, Shujie Chen, Liangjing Wang, Microbial metabolite enhances immunotherapy efficacy by modulating T cell stemness in pan-cancer, Cell, Volume 187, Issue 7, 2024, Pages 1651-1665.e21, ISSN 0092-8674, https://doi.org/10.1016/j.cell.2024.02.022. (https://www.sciencedirect.com/science/article/pii/S0092867424002241)
6. Kreiter, S., Vormehr, M., van de Roemer, N. et al. Mutant MHC class II epitopes drive therapeutic immune responses to cancer. Nature 520, 692–696 (2015). https://doi.org/10.1038/nature14426
7. Huang, F., Li, S., Chen, W., Han, Y., Yao, Y., Yang, L., Li, Q., Xiao, Q., Wei, J., Liu, Z., Chen, T., & Deng, X. (2023). Postoperative Probiotics Administration Attenuates Gastrointestinal Complications and Gut Microbiota Dysbiosis Caused by Chemotherapy in Colorectal Cancer Patients. Nutrients, 15(2), 356. https://doi.org/10.3390/nu15020356
8. Hundal, J., Kiwala, S., McMichael, J., Miller, C. A., Xia, H., Wollam, A. T., Liu, C. J., Zhao, S., Feng, Y. Y., Graubert, A. P., Wollam, A. Z., Neichin, J., Neveau, M., Walker, J., Gillanders, W. E., Mardis, E. R., Griffith, O. L., & Griffith, M. (2020). pVACtools: A Computational Toolkit to Identify and Visualize Cancer Neoantigens. Cancer immunology research, 8(3), 409–420. https://doi.org/10.1158/2326-6066.CIR-19-0401

9. Yang, K., Halima, A. & Chan, T.A. Antigen presentation in cancer — mechanisms and clinical implications for immunotherapy. Nat Rev Clin Oncol 20, 604–623 (2023). https://doi.org/10.1038/s41571-023-00789-4

10. Kruse, B., Buzzai, A. C., Shridhar, N., Braun, A. D., Gellert, S., Knauth, K., Pozniak, J., Peters, J., Dittmann, P., Mengoni, M., van der Sluis, T. C., Höhn, S., Antoranz, A., Krone, A., Fu, Y., Yu, D., Essand, M., Geffers, R., Mougiakakos, D., Kahlfuß, S., … Tüting, T. (2023). [CD4+ T cell-induced inflammatory cell death controls immune-evasive tumours. Nature, 618(7967), 1033–1040. https://doi.org/10.1038/s41586-023-06199-x

11. Henrickson, S., Mempel, T., Mazo, I. et al. T cell sensing of antigen dose governs interactive behavior with dendritic cells and sets a threshold for T cell activation. Nat Immunol 9, 282–291 (2008). https://doi.org/10.1038/ni1559

12. Wiertsema, S. P., van Bergenhenegouwen, J., Garssen, J., & Knippels, L. M. J. (2021). The Interplay between the Gut Microbiome and the Immune System in the Context of Infectious Diseases throughout Life and the Role of Nutrition in Optimizing Treatment Strategies. Nutrients, 13(3), 886. https://doi.org/10.3390/nu13030886

13. Guo, Y., Gao, M., Wang, L., Yuan, H., Yin, J., Wu, J., Gao, X., Zhu, Z., Zhang, Y., Wang, Z., Huang, H., & Kang, G. (2025). An Engineered Probiotic Consortium Based on Quorum-Sensing for Colorectal Cancer Immunotherapy. Advanced science (Weinheim, Baden-Wurttemberg, Germany), 12(46), e12744. https://doi.org/10.1002/advs.202512744

14. Huang, F., Li, S., Chen, W., Han, Y., Yao, Y., Yang, L., Li, Q., Xiao, Q., Wei, J., Liu, Z., Chen, T., & Deng, X. (2023). Postoperative Probiotics Administration Attenuates Gastrointestinal Complications and Gut Microbiota Dysbiosis Caused by Chemotherapy in Colorectal Cancer Patients. Nutrients, 15(2), 356. https://doi.org/10.3390/nu15020356